

CUSTOMER FEATURE GUIDE

Atlas

The central storage control plane for the Zyvor
product suite.

One intent-based API that maps every product's storage request to Ceph, NFS, ZFS, or the cloud — while owning inventory, ownership, protection, and audit.

by ZyvorAI Labs

What is Atlas?

Atlas lets products ask for intent — "give me production block storage" — instead of wrestling with pool internals, then maps that intent to a real backend through pluggable drivers. It gives you block, file, and object storage from a single gateway, with async provisioning, snapshots, backups, replication, per-tenant governance, and a live console on top. Ceph is the first driver (RBD, CephFS, RGW/S3), with NFS and ZFS backends alongside it and DataBridge adding cloud-to-edge database and object mobility.

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Storage backends
(Ceph · NFS · ZFS)

6

Database engines
migratable via
DataBridge

80+

REST endpoints across
the control plane

3

Access surfaces —
REST · gRPC · SSE

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Control Plane & Architecture

An intent-driven gateway that decouples every product from the storage underneath it.



Intent-Based Provisioning

Products request an intent class ("production block storage") and Atlas resolves it to a concrete backend, pool, and placement.

Callers stay decoupled from Ceph or any future backend — no pool internals leak into product code.



Pluggable Storage Drivers

A single StorageDriver trait fronts every backend, with real Ceph, NFS, ZFS, and Kubernetes drivers plus a fake driver for local runs.

Add a backend once and every product gets it through the same stable API.



Atlas Gateway

An axum server that centralizes auth, audit, and the API surface, and embeds the Storage Center console in the binary.

One deployable front door for storage across all nine Zyvor products.



REST + gRPC Surfaces

The same control plane is reachable over REST and a tonic gRPC edge with streaming WatchJob for product integrations.

Human tools use REST; product runtimes get typed, streaming gRPC.



Inventory & Discovery

A discovery worker normalizes backend state into a SQLite inventory of pools, volumes, and ownership bindings.

A consistent, queryable picture of every asset regardless of which backend holds it.



Durable, Self-Healing State

Restarts recover the job engine (interrupted jobs fail safe, queued jobs re-enqueue), with graceful SIGTERM drain and deep readiness probes.

The control plane stays trustworthy across pod restarts and rollouts.

Core principle: products request **intent**, not pool internals. Atlas maps intent to a backend and owns inventory, ownership, and audit.

Block, File & Object Storage

Provision and manage RBD block, CephFS file, and RGW/S3 object storage from one API.



Block Volumes

Create, get, expand, and delete Ceph-backed block volumes (PVCs) as async jobs that return a job id immediately.

[Fast, non-blocking volume lifecycle with a durable audit of every change.](#)



CephFS File Shares (RWX)

Provision shared read-write-many file storage on CephFS for workloads that need concurrent access.

[Multi-writer file storage without standing up a separate NAS.](#)



Object Buckets

Create S3 buckets via ObjectBucketClaim with per-bucket quotas, stats, and presigned upload/download URLs.

[Self-service object storage with quota control and short-lived access links.](#)



Direct RBD Image Ops

Provision, clone, resize, flatten, snapshot, and roll back RBD images directly, with per-image usage tracking.

[Full low-level control when you need to bypass the PVC abstraction.](#)



Safe Resize (Grow & Shrink)

Expand volumes freely; shrink is opt-in behind an explicit allow_shrink flag as a data-loss guard.

[Reclaim over-provisioned space without accidentally destroying data.](#)



Kubernetes Storage Inventory

Read-only listing of StorageClasses, PVCs, and PVs live from the cluster via kube-rs.

[See exactly how Atlas storage surfaces inside Kubernetes.](#)

Data Protection

Snapshots, clones, and off-cluster backups — scheduled, verified, and retention-managed.



Snapshots

Take point-in-time volume snapshots, then clone or restore from them behind a safe-delete guard.

Instant rollback points without copying whole volumes.



Clone & Restore

Spin a new volume from a snapshot or restore a volume in place from any snapshot.

Branch environments from a known-good state in seconds.



Off-Cluster Backups

Stream RBD export-diff to RGW/S3 as verified multipart uploads, with restore-from-data and presigned downloads.

Durable, verifiable backups that survive loss of the source pool.



Scheduled Snapshots & Backups

Register recurring snapshot and backup schedules per volume, driven by background workers.

Set-and-forget protection instead of manual, forgettable runs.



Retention Policies

Keep-N plus max-age retention automatically prunes old backups (and object versions).

Bounded storage cost without hand-pruning stale copies.



Orphan Backup GC

Surface backups whose source volume is gone so they can be cleaned up deliberately.

No silent buildup of orphaned data you're still paying to store.

Multi-Backend & Ceph Operations

Run Ceph, NFS, and ZFS side by side — and operate Ceph natively when it's the driver.



Three Storage Backends

Ceph, NFS, and ZFS all live behind the same StorageDriver, with filters, per-backend gauges, and a backends summary.

[Mix backends under one control plane and one API.](#)



Dynamic Backend Registration

Register a live NFS or ZFS backend over the API and Atlas instantiates the driver and discovers it immediately.

[Onboard new storage without redeploying the gateway.](#)



Backend Cordon & Drain

Cordon a backend to reject new provisioning (503) while you drain and service it, then uncordon.

[Take storage offline for maintenance without breaking callers.](#)



Ceph-Native Introspection

Read ceph status, osd-tree, osd-df, and df directly through dedicated endpoints.

[Diagnose the real cluster without leaving the control plane.](#)



OSD Management

Mark OSDs in or out and reweight them via ceph osd commands from the API.

[Rebalance and service OSDs through the same audited surface.](#)



QoS & Pool Migration

Set per-image IOPS/BPS QoS limits and migrate images between pools with rbd migration.

[Tune performance and rebalance placement without downtime.](#)

OSD, QoS, and pool-migration operations act on a **real Ceph cluster** — they are exercised fake-first in CI and fully verified against live Rook Ceph.

Observability & Metrics

Live capacity, health, forecasts, and a cinematic console view of the whole estate.



Prometheus Metrics

Self-metrics at /metrics plus Ceph metrics and per-backend gauges for scraping.

[Drop Atlas straight into your existing monitoring stack.](#)



Persisted Time-Series History

Capacity and usage history is persisted and served at /metrics/history for trend analysis.

[See where storage has been, not just where it is now.](#)



Capacity Forecast

A days-to-full forecast projects when pools and backends will run out of headroom.

[Buy or reclaim capacity before you hit the wall.](#)



Observatory

Six live canvas visualizations render the cluster, pools, jobs, and flows in real time in the console.

[An at-a-glance operations view that reads like mission control.](#)



Unified Activity Feed

A single /events stream merges activity across every backend and subsystem.

[One timeline for everything happening in storage.](#)



Grafana Bundle

A ready-to-apply Prometheus + Grafana observability bundle ships in deploy/observability.

[Stand up dashboards in minutes instead of building them from scratch.](#)

Alerting & Day-2 Ops

Atlas is operable, not just observe-and-provision — with alerts, maintenance, and safe upgrades.



Rule-Based Alerting

Rules cover cluster health, pool near-full, OSD down/latency, capacity forecast, recovery, failing jobs, CDC errors, and tenant-quota thresholds.

[Catch storage problems by policy, not by luck.](#)



Alert Lifecycle

Acknowledge, silence for a window, or resolve alerts, with a single webhook sink for downstream notification.

[Route real signal to your team and mute the noise.](#)



Maintenance Pause

Pause the job worker so new work holds in queue until you resume — cluster-wide freeze for maintenance.

[Do risky work with nothing new landing mid-flight.](#)



Upgrade Pre-Flight

A preflight check blocks upgrades on HEALTH_ERR clusters, open critical alerts, in-flight jobs, or lagging CDC.

[Never ship an upgrade into an unhealthy cluster.](#)



Gated Rollout & Rollback

deploy-remote.sh auto-gates on pre-flight and supports rollout undo, with a --force override.

[Ship and un-ship with guardrails baked into the script.](#)



Control-Plane State Backup

Optional VACUUM INTO snapshots of the control-plane DB stream to S3/RGW on a schedule with keep-N retention.

[Recover the control plane itself, not just the data it manages.](#)

Governance & Multi-Tenancy

Tenant isolation, quotas, policy, audit, and cost attribution — built into the control plane.



Multi-Tenancy

Per-tenant quotas and policy overrides isolate consumers of shared storage.

Give each team its own guardrails on one physical cluster.



Tenant Quotas

Set and read capacity quotas per tenant, with alerts as usage approaches 80% and 95%.

Prevent one tenant from starving everyone else.



Placement Policies & Drift

Intent-to-placement policies govern where storage lands, and a policy-drift report flags volumes that no longer match.

Keep placement compliant with intent over time, not just at creation.



Service-Account JWTs & RBAC

Issue role-scoped JWTs and revoke leaked tokens before their TTL, enforced by RBAC across REST and gRPC.

Grant least-privilege access and kill compromised tokens instantly.



Audit Log

Every mutating action is recorded, exportable as CSV, with configurable retention-day pruning.

A defensible trail of who changed what, when.



Chargeback

Attribute cost per tenant from a configurable USD-per-GiB-month rate.

Turn shared storage into a billable, accountable service.

Governance controls include a per-minute **rate limiter** (429 on breach) so a runaway caller can't overwhelm the gateway.

DataBridge — Database Mobility

Migrate managed cloud databases to open, self-managed engines on Ceph at the edge.

Source engine	Edge target	Migration type	Real connector
PostgreSQL	CloudNativePG	homogeneous	default (tokio-postgres)
MySQL / MariaDB	Percona XtraDB	homogeneous	default (sqlx)
Oracle	CloudNativePG (Postgres)	heterogeneous	feature: oracle (OCI)
SQL Server	CloudNativePG (Postgres)	heterogeneous	feature: sqlserver
MongoDB	Percona Server for MongoDB	homogeneous (document)	feature: mongoddb



Discover & Assess

Introspect a live source database (tables, PKs, sizes, WAL/CDC capability) and score edge-readiness.

[Know exactly what you're migrating — and its risks — before you start.](#)



Edge Provisioning on Ceph

Provision CloudNativePG, Percona XtraDB, or Percona Server for MongoDB with data and WAL on Ceph RBD.

[Land your database on open engines and your own storage, off the cloud bill.](#)



Full-Load + CDC Replication

Dump-and-restore full load, then Debezium streams ongoing changes (WAL/binlog/redo/oplog) via Kafka to the edge.

[Migrate with the source still live — no big-bang downtime.](#)



Guarded Cutover & Rollback

Cutover is admin-guarded on validation-passed and low CDC lag, with rollback allowed inside a defined window.

[Switch over safely and back out if something looks wrong.](#)



Six Source Engines

PostgreSQL, MySQL, MariaDB, Oracle, SQL Server, and MongoDB — homogeneous and heterogeneous (to Postgres) paths.

[One migration control plane spanning your whole database estate.](#)



CDC Self-Heal

A stalled stream is re-established via cdc/restart, and the reconciler auto-restarts an unhealthy stream up to 3x before alerting.

[Replication recovers itself instead of quietly falling behind.](#)

DataBridge — Object Migration

Move AI datasets, model weights, and RAG documents from cloud object stores into Ceph RGW.



Cloud-to-Ceph Object Copy

Copy objects from a cloud store into a Ceph RGW bucket so products consume from a local S3 endpoint.

[Pull AI-scale datasets on-prem and off the cloud egress meter.](#)



Incremental, Verified Copy

List source, diff against destination by key+size, stream each changed object recording sha256, then verify every key landed.

[Trustworthy syncs that only move what changed and prove they arrived.](#)



Multi-Cloud Sources

AWS S3, GCS (S3-interop), and any S3-compatible store work today; Azure Blob has a native feature-gated connector.

[One mover for the object stores you actually use.](#)



Concurrent Streaming

Objects stream and copy concurrently with configurable part size and concurrency for AI-scale datasets.

[Move terabytes fast instead of one slow object at a time.](#)



Secret-Ref Credentials

Credentials are referenced from a Kubernetes Secret, resolved in-process at run time, and never stored or logged.

[Migrate data without spilling access keys into the control plane.](#)

Disaster Recovery

Cross-cluster RBD mirroring and failover scaffolding for a second Ceph site.



Cluster Peering

Register and remove peer Ceph clusters as DR targets through the control plane.

[Catalog your DR relationships in one place.](#)



Volume Mirroring

Enable or disable rbd mirror on a volume and list all mirrors with their DR status.

[Keep critical volumes replicated to a second site.](#)



Promote & Demote (Failover)

Promote a mirror at the DR site or demote the primary to orchestrate failover.

[Flip sites when the primary goes down.](#)

Honest status: the DR **control-plane catalog, API, and jobs are complete**, but the underlying **rbd mirror operations are not yet verified against a live second Ceph cluster** — treat DR as scaffolding pending a two-site validation.

Storage Center Console

A Zeus OS-style React console embedded in the gateway, wiring every capability to a UI.



Embedded React Console

A React/Vite/Tailwind SPA served from the gateway binary over HTTPS with a branded login.

Nothing extra to deploy — the UI ships inside Atlas.



Full-Capability Coverage

Inventory, capacity/health, job progress via SSE, alerts, metrics, tenants, and every write action are wired into the UI.

Drive the whole control plane without touching the CLI.



Live Job Progress (SSE)

Async jobs stream progress to the console over server-sent events as they run.

Watch provisioning, backups, and migrations advance in real time.



Themeable Design

A Nebula default theme plus Midnight and Aurora variants in the Zeus OS "Tahoe" design language.

A console that looks like the rest of the Zyor suite.



atlasctl CLI

A headless REST client covering health, discovery, volumes, snapshots, backups, buckets, RBD, tenants, tokens, and more.

Script and automate everything the console can do.

GETTING STARTED

Your first 30 minutes

A suggested path to value on day one.

- 1 Run the gateway (no Ceph, no cluster)**
make run starts the gateway with the fake Ceph driver on 127.0.0.1:5110 — no real storage required.
- 2 Populate inventory**
cargo run -p atlas-cli -- discover pulls a normalized inventory from the driver; then pools, volumes, and health are live.
- 3 Open the Storage Center**
Browse to <http://127.0.0.1:5110/> for the embedded console — inventory, capacity, jobs, alerts, tenants, and write actions.
- 4 Try DataBridge fake-first**
make run-databridge runs the whole discover-to-cutover pipeline with no cloud or Kubernetes creds needed.
- 5 Go real on k3s + Rook Ceph**
Use `deploy/rook-ceph-lab` and `scripts/deploy-remote.sh` to stand up real Ceph and deploy the gateway in Ceph mode.

Support & contact

Atlas is developed by **ZyvorAI Labs**. For onboarding, a proof-of-concept, or a guided demo, contact your account team at info@zyvor.dev.

Good to know: Atlas is at slices 1–5 plus DataBridge, verified end-to-end on real k3s + Rook Ceph clusters. Some capabilities require real infrastructure or optional build features: cross-cluster DR (rbd mirror) is complete as control-plane scaffolding but unverified without a live second Ceph cluster; true multi-replica HA requires Postgres (single-replica SQLite is the default); Oracle, SQL Server, MongoDB, precise CDC lag, and Azure Blob connectors are behind cargo features that link native dependencies; and OSD/QoS/pool-migration and live CDC for non-Postgres engines exercise real Ceph and a running Kafka/Debezium stack respectively.

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